Detecting Cytokine Cascade Direction from Latent Geometry

Full Method, Validation, and Refinement

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1 Biological Motivation

Cytokine cascades are directional: cytokine A activates cells of type X, which produce cytokine B as a downstream effector. The goal is to infer these directed relationships — and identify the relay cell type X — from single-cell RNA-seq data, without prior knowledge of which cytokines cascade into which.

The key observation is the following. In the Oesinghaus et al. dataset, cells are cultured in pseudo-tubes: closed bags of $\approx$480 PBMCs treated with a single cytokine for 24 hours. If cascade $A \to B$ exists, cells inside an A-treated tube will have begun responding to *endogenously produced* B — because cells that were directly activated by A start secreting B into the same tube. As a result, an A-tube contains a mixed signal: a strong direct response to A and a weaker secondary response to B.

Crucially, the reverse is not true: a B-treated tube contains no endogenously produced A. This **directional asymmetry** — A-tubes look partially like B-tubes, but not vice versa — is the cascade signal we aim to detect.

2 Mathematical Setting

Cells and expression. Let $\mathcal{I}$ be the set of all cells. Each cell $i \in \mathcal{I}$ has a gene expression vector $x_i \in \mathbb{R}^G$ (log-normalised, $G = 4{,}000$ HVGs), a cytokine label $c(i) \in \mathcal{C}$ (where C_0 denotes PBS), a cell-type label $\tau(i) \in \mathcal{T}$ ($M = 18$ annotated PBMC types), and a donor label $d(i) \in \mathcal{D}$ ($|\mathcal{D}| = 12$).

Encoder. A residual MLP $f_\theta : \mathbb{R}^G \to \mathbb{R}^d$ ($d = 128$) maps each cell to a dense embedding $h_i = f_\theta(x_i)$. f_θ is pre-trained by cell-type classification (Stage 1) and then frozen; it has never seen cytokine labels, so all cytokine signal in h_i is emergent.

Donors. The dataset is split at the donor level: $\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}}$, with $\mathcal{D}_{\text{train}} = \{\text{D1, D4–D12}\}$ and $\mathcal{D}_{\text{val}} = \{\text{D2, D3}\}$. All centroids are computed on $\mathcal{D}_{\text{train}}$ only.

Cell sets. For cytokine $A \in \mathcal{C}$ and cell type $T \in \mathcal{T}$: $\mathcal{S}_{A,T} = \{i \in \mathcal{I} : c(i) = A,\ \tau(i) = T,\ d(i) \in \mathcal{D}_{\text{train}}\}$.

3 The PBS-Relative Centroid (PBS-RC) Transform

3.1 The Composition Problem

The dominant axis of variation in $\mathbb{R}^d$ is cell type: every NK cell lands near the NK cluster regardless of cytokine treatment. A naïve comparison of cytokine centroids $\bar{h}_A = |\mathcal{S}_A|^{-1} \sum_{i \in \mathcal{S}_A} h_i$ conflates two unrelated sources of variation: (1) *cell-state changes* induced by the cytokine, and (2) *cell-type composition* differences between conditions (e.g. if A expands NK cells, $\bar{h}_A$ shifts toward the NK cluster even if no NK cell changed its transcriptional state).

One might think that computing the within-tube displacement $(\boldsymbol{\mu}_{A,T} - \boldsymbol{\mu}_A)$ for a fixed cell type T already avoids this problem. It does not. If cytokine A causes NK cell proliferation, there are proportionally more NK cells in A-tubes, which pulls $\boldsymbol{\mu}_A$ toward the NK cluster. This shifts $(\boldsymbol{\mu}_{A,\text{NK}} - \boldsymbol{\mu}_A)$ in the direction of NK space even if NK cells are transcriptionally identical to their resting state — producing spurious bias. The PBS-RC transform resolves this: by subtracting the resting-state reference per cell type, $\tilde{h}_i = \mathbf{0}$ for any cell that has not changed state from PBS. Proliferation alone does not displace centroids in PBS-RC space.

3.2 PBS-RC Embedding

Definition 1 (PBS reference centroid)**.** *For each cell type $T \in \mathcal{T}$, the PBS reference centroid is*

$$\boldsymbol{\mu}_{\mathrm{PBS},T} = \frac{1}{|\mathcal{S}_{C_0,T}|} \sum_{i \in \mathcal{S}_{C_0,T}} h_i \in \mathbb{R}^d.$$

Definition 2 (PBS-RC embedding)**.** *The PBS-Relative Centroid embedding of cell i is $\tilde{h}_i = h_i - \boldsymbol{\mu}_{\mathrm{PBS},\tau(i)} \in \mathbb{R}^d$.*

By construction, $\tilde{h}_i = \mathbf{0}$ whenever cell i occupies the same position in embedding space as the average resting-state cell of type $\tau(i)$. A non-zero $\tilde{h}_i$ represents a deviation from resting state for that specific cell type. Because the subtraction is cell-type-specific, it genuinely reshapes the geometry rather than shifting all cells by the same offset.

3.3 Cytokine Centroids

For cytokine $A \in \mathcal{C} \setminus \{C_0\}$, define the cytokine centroid and the cell-type-specific mean in PBS-RC space:

$$\boldsymbol{\mu}_A = \frac{1}{|\mathcal{S}_A|} \sum_{i \in \mathcal{S}_A} \tilde{h}_i, \qquad \boldsymbol{\mu}_{A,T} = \frac{1}{|\mathcal{S}_{A,T}|} \sum_{i \in \mathcal{S}_{A,T}} \tilde{h}_i.$$

The deviation $(\boldsymbol{\mu}_{A,T} - \boldsymbol{\mu}_A)$ measures how much cell type T is displaced within A-tubes relative to the average A-tube cell.

Figure 1 illustrates the transform: in raw embedding space cell-type identity dominates; after PBS-RC, all cell types are centred at their own resting state and cytokine-induced displacements become visible.

4 Cascade Direction Scores: Legacy Method

4.1 Directional Bias

For cytokines $A, B \in \mathcal{C} \setminus \{C_0\}$ with $A \neq B$, let

$$\mathbf{u}_{A \to B} = \frac{\boldsymbol{\mu}_B - \boldsymbol{\mu}_A}{\|\boldsymbol{\mu}_B - \boldsymbol{\mu}_A\|_2}$$

be the unit vector pointing from A's centroid to B's centroid. The **directional bias** of cell type T in A-tubes toward B is:

$$\mathrm{bias}(A \to B, T) = \big(\boldsymbol{\mu}_{A,T} - \boldsymbol{\mu}_A\big) \cdot \mathbf{u}_{A \to B} \in \mathbb{R}.$$

The two components work together. $(\boldsymbol{\mu}_{A,T} - \boldsymbol{\mu}_A)$ is cell type T's displacement relative to the average A-tube cell — it is non-zero only when T stands out from the rest of the tube. Without this subtraction, $\boldsymbol{\mu}_{A,T}$ would reflect cell-type identity rather than cytokine response. $\mathbf{u}_{A \to B}$ is the direction cells move when exposed to B: because the encoder was trained on all conditions simultaneously, the $A \to B$ axis is exactly the direction of secondary B-exposure. Projecting onto it is therefore not an arbitrary choice — in the encoder's geometry, "experiencing B" and "moving in the $\mathbf{u}_{A \to B}$ direction" are the same thing.

Crucially, $\mathbf{u}_{A \to B}$ is the same for all cell types, so bias values for different T are measured on the same axis and directly comparable.

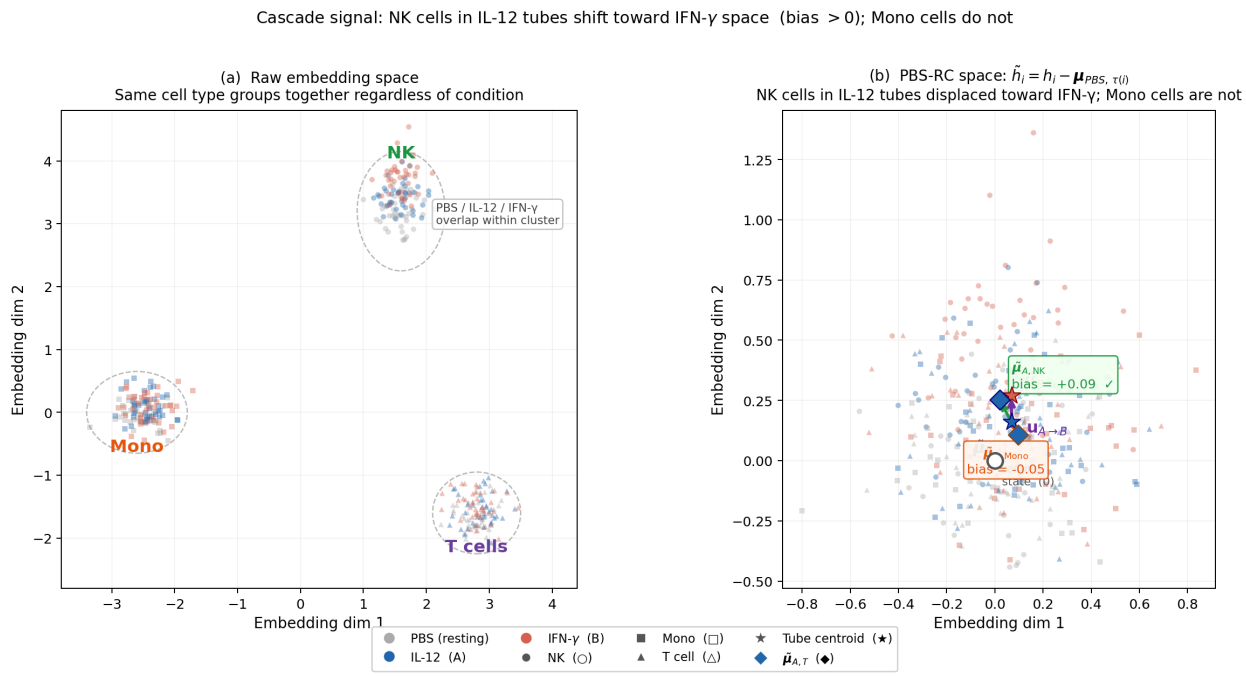

Figure 1: **PBS-RC reveals within-cluster cytokine displacements.** *Left:* Raw encoder space. Cell-type identity dominates; IL-12 and IFN-γ tube centroids nearly coincide, and the NK displacement $\boldsymbol{\mu}_{A,\mathrm{NK}} - \boldsymbol{\mu}_A$ spans the whole space. *Right:* After the PBS-RC transform $\tilde{h}_i = h_i - \boldsymbol{\mu}_{\mathrm{PBS},\tau(i)}$, all cell types are centred at the origin, cytokine centroids separate, and NK cells in IL-12 tubes (NK cascade relay) are visibly displaced toward IFN-γ space (bias $= +0.09$), while Monocytes show no displacement (bias $= -0.05$).

4.2 Why This Captures Cascade Direction

Suppose cascade $A \to B$ exists via relay cell type X:

1. Cytokine A is added to the pseudo-tube.
2. Cells of type X are primary responders: they are directly activated by A and begin secreting B into the tube. X is therefore exposed to B autocrinally — its transcriptional state is shaped by *both* A and B.
3. All other cells in the tube receive B paracrinally and shift toward B-space.

X receives a double signal while bystander cells receive only the paracrine B signal. X therefore shifts *more* than the average tube cell toward B-space: $\mathrm{bias}(A \to B, X) > 0$. In a B-treated tube there is no endogenously produced A; X responds to B just like every other cell type, so $\mathrm{bias}(B \to A, X) \approx 0$.

If A and B share a pathway without either producing the other (e.g. IL-6 and IL-10 via STAT3), both biases are small and symmetric, so the method correctly returns a near-zero score for this negative control.

4.3 Legacy Asymmetry Score and Its Two Problems

The legacy aggregate score was:

$$\mathrm{ASYM}(A \to B) = \max_{T \in \mathcal{T}} \big[\mathrm{bias}(A \to B, T) - \mathrm{bias}(B \to A, T)\big].$$

The maximum over cell types is motivated by the fact that a cascade acts through a specific relay; averaging over all $M-1$ bystander cell types would dilute the signal. The arg max directly identifies the relay: $X^* = \arg\max_T \mathrm{asym}(A \to B, T)$.

Despite being well-motivated, the subtraction inside the max introduces two structural defects.

Contamination. Since $\hat{u}_{B \to A} = -\hat{u}_{A \to B}$, expanding the subtraction gives:

$$\mathrm{bias}(A \to B, T) - \mathrm{bias}(B \to A, T) = \underbrace{(\boldsymbol{\mu}_{A,T} - \boldsymbol{\mu}_A) \cdot \hat{u}_{A \to B}}_{\text{forward (wanted)}} + \underbrace{(\boldsymbol{\mu}_{B,T} - \boldsymbol{\mu}_B) \cdot \hat{u}_{A \to B}}_{\text{contaminant}}.$$

The second term is the projection of B's own cell-type deviation onto the $A \to B$ direction. A cell type T that responds strongly *directly* to B inflates $\mathrm{ASYM}(A \to B)$ even when no cascade from A through T exists.

Forced antisymmetry. Because $b_{\mathrm{fwd}} - b_{\mathrm{rev}}$ and $b_{\mathrm{rev}} - b_{\mathrm{fwd}}$ differ only in sign, the score satisfies $\mathrm{ASYM}(A \to B) = -\mathrm{ASYM}(B \to A)$ identically. A finding $A \to B$ and a finding $B \to A$ are algebraically equivalent; the method cannot distinguish a genuine directed cascade from a sign flip.

Figure 2 illustrates the full method (panels A–D), including panel D which shows explicitly why the naïve alternative formula $\boldsymbol{\mu}_{A,T} - \boldsymbol{\mu}_{B,T}$ also fails: it detects differential response rather than convergent cascade response, inverting the cell-type ranking relative to the ground truth.

5 Refined Pipeline

The refined pipeline keeps the same geometric intuition — cell-type embeddings deviate toward cascade targets in the latent space — but replaces the contaminated subtraction with two fully independent one-sided statistical tests.

5.1 Step 1 — Per-Donor PBS-RC Space

For every cell type T and every training donor d, compute the PBS centroid:

$$\boldsymbol{\mu}^{(d)}_{\mathrm{PBS},T} = \frac{1}{|S|} \sum_{i \in S} h_i, \quad S = \{\text{cells of type } T \text{ from donor } d \text{ under PBS}\}.$$

Subtract per cell: $\tilde{h}_i = h_i - \boldsymbol{\mu}^{(d_i)}_{\mathrm{PBS},\tau(i)}$. Cytokine-specific centroids $\tilde{\boldsymbol{\mu}}^{(d)}_{C,T}$ now represent the deviation of cell type T from its own resting state under cytokine C, keeping comparisons on the same baseline regardless of donor-specific cell-type composition.

5.2 Step 2 — Per-Donor Forward Projection with Centroid Subtraction

For an ordered pair (A, B) and cell type T, the forward bias for donor d is:

$$b_{\text{fwd}}^{(d)}(A \to B, T) = \left(\tilde{\boldsymbol{\mu}}_{A,T}^{(d)} - \tilde{\boldsymbol{\mu}}_A\right) \cdot \hat{u}_{A \to B},$$

where $\tilde{\boldsymbol{\mu}}_A = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_d \tilde{\boldsymbol{\mu}}_A^{(d)}$ is cytokine A's pooled centroid and $\hat{u}_{A \to B}$ is the direction between pooled centroids. Subtracting $\tilde{\boldsymbol{\mu}}_A$ removes the component of A's global response shared across all cell types, so the score captures T's *specific* deviation toward B beyond A's generic direct effect. No reverse term is subtracted anywhere; the "reverse" test for (A, B) is simply the forward test for (B, A), computed independently.

5.3 Step 3 — Statistical Testing

For each ordered pair (A, B) and cell type T, a one-sided Wilcoxon signed-rank test is applied to $\{b_{\text{fwd}}^{(d)}(A \to B, T)\}_{d \in \mathcal{D}_{\text{train}}}$ against the null of zero median (H_1: median > 0). Multiple testing is handled in two stages:

1. **Bonferroni** across cell types within each ordered pair (relay search: find the specific cell type driving the signal).
2. **Benjamini–Hochberg FDR** ($\alpha = 0.05$) across all $K(K-1) = 8{,}010$ ordered pairs.

The cascade call for pair (A, B) is determined by:

$$\text{fwd_sig} = \exists\, T : p_{\text{Bonf}}(A \to B, T) \leq \alpha, \qquad \text{rev_sig} = \exists\, T : p_{\text{Bonf}}(B \to A, T) \leq \alpha,$$

producing one of four labels: `A->B`, `B->A`, `shared` (both significant), or `none`.

6 Experimental Setup

6.1 Batch Construction

The full dataset contains 12 donors. For each cytokine c_i (sorted alphabetically, $i = 0, \ldots, 90$), one tube is selected from donor $(i + \delta) \bmod 12$, where δ is the donor offset. Two mutually exclusive batches were constructed:

- **Batch 0** ($\delta = 0$, seeds 1–10, 42, 123): cytokine 0 from donor 0, cytokine 1 from donor 1, ...
- **Batch 1** ($\delta = 1$, seeds 11–22): cytokine 0 from donor 1, cytokine 1 from donor 2, ...

Every cytokine is represented by a *different* donor across the two batches, so the two sets of training cells are mutually exclusive. Within each batch, the 12 seeds vary only the random weight initialisation.

6.2 Pre-Registered Controls

Before inspecting any results, four cytokine pairs were designated as controls (Table 1). IL-12 $\to$ IFN-γ is the best-documented PBMC cascade in the dataset (IL-12 activates NK cells, which produce IFN-γ; Oesinghaus et al. Fig. 4i). IL-6 and IL-10 share the STAT3 signalling axis but have no documented directional cascade, so the signal is expected to be symmetric.

Table 1: Pre-registered pairs of interest.

Direction	Pair	Role	Expected signal
$A \to B$	IL-12 $\to$ IFN-γ	Positive control	High score; $A \to B > B \to A$
$B \to A$	IFN-γ $\to$ IL-12	Positive control (reverse)	Lower score than forward
$A \to B$	IL-6 $\to$ IL-10	Negative control	Mid-range, symmetric with reverse
$B \to A$	IL-10 $\to$ IL-6	Negative control (reverse)	Similar rank to forward

7 Results

7.1 Legacy Method

Within-batch seed stability. A seed is classified as *stable* if its Spearman ρ with the batch ensemble mean (8,010-pair vector) is ≥ 0.70. The legacy method achieves 10/12 stable seeds in Batch 0 and 11/12 in Batch 1.

Cross-batch reproducibility. The two stable ensembles are computed from disjoint training cells. Their Spearman correlation is:

$$\rho\big(\text{Batch-0 ensemble, Batch-1 ensemble}\big) = 0.766 \quad (p < 10^{-300}),$$

exceeding the pre-specified threshold of 0.70. In addition, 22 of the top-50 pairs (44%) appear in the top-50 of both batches.

Pre-registered controls. The positive control IL-12 $\to$ IFN-γ ranks in the top 9–13% in both batches (Table 2), consistently higher than its reverse (top 16–18%) — directional asymmetry in the expected direction. The negative control IL-6/IL-10 ranks in the middle of the distribution and the two directions are nearly symmetric (rank difference ≤ 600), consistent with a shared-pathway signal.

Table 2: Pre-registered pair ranks under the legacy method.

Source	Target	Role	B0 Asym	B0 Rank	B1 Rank
IL-12	IFN-γ	pos. ctrl.	3.025	699/8010	1007/8010
IFN-γ	IL-12	pos. ctrl. (rev.)	2.527	1424/8010	1261/8010
IL-6	IL-10	neg. ctrl.	1.640	3828/8010	3348/8010
IL-10	IL-6	neg. ctrl. (rev.)	1.944	2839/8010	2796/8010

Top discovered pairs. Two cytokines dominate the top of both batch rankings: **IL-27** (a Th1-promoting cytokine with documented downstream targets including IL-22, TNF-α, and proliferation-associated ligands) and **IFN-β** (a type-I interferon with documented downstream activation of ISG programs and secondary cytokine production). The 10 pairs with the highest mean asymmetry score across stable seeds are shown in Figure 3; their consistent appearance in both disjoint batches is evidence that the signal is biological rather than a data artefact.

7.2 Refined Method vs. Legacy: Comparison

Both methods were re-evaluated on the full 24-run corpus (Batch 0 and Batch 1, 12 seeds each). Figure 4 shows per-seed ρ for both methods side by side.

Seed stability. The refined method improves within-batch stability from 7/12 to 11/12 in Batch 0, while leaving Batch 1 unchanged. Seed 123 is the sole persistent outlier for the refined method across both batches.

Cross-batch reproducibility and the filter test. Table 3 reports cross-batch Spearman ρ and top-50 pair overlap under four pair-set filters. The overall refined ρ (0.815) exceeds legacy (0.772). The most diagnostic row is the third: when restricting to the top 50% of pairs *by legacy magnitude*, the legacy ρ *drops* to 0.620 while the refined ρ on the same pairs *rises* to 0.867. This means the legacy method's highest-magnitude pairs are its *least* reproducible — the contamination term $(\boldsymbol{\mu}_{B,T} - \boldsymbol{\mu}_B) \cdot \hat{u}_{A\to B}$ inflates scores for strong direct-B responders without adding biological signal.

Conversely, the refined ρ improves monotonically with filtering (rows 1–3: $0.815 \to 0.835 \to 0.867$), indicating that the refined score does carry a real signal that concentrates toward the top of its own ranking.

The top-50 overlap for the refined method is still modest (3–10/50), indicating that a noise floor persists at the level of individual pair rankings. However, the monotone improvement with filtering suggests this is a *threshold* problem, not a signal problem: applying a stricter BH threshold or using more seeds should push overlap higher.

Table 3: Cross-batch Spearman ρ and top-50 overlap under four pair-set filters. "Top-50 overlap" counts how many of the top-50 pairs by Batch-0 score also appear in the top-50 of Batch-1. Row 3 is the critical diagnostic: legacy ρ drops when restricted to its own high-magnitude pairs; refined rises.

Filter	**Legacy ρ**	**Refined ρ**	**Legacy overlap**	**Refined overlap**
All pairs ($n = 8{,}010$)	0.772	0.815	25/50	3/50
Top 50% by refined score ($n = 4{,}005$)	0.762	0.835	28/50	10/50
Top 50% by |legacy| score ($n = 4{,}005$)	0.620	0.867	25/50	9/50
Active pairs (refined > 0, $n = 7{,}100$)	0.772	0.813	27/50	3/50

Pre-registered controls under the refined method. Neither method confidently calls the pre-registered IL-12 $\to$ IFN-γ direction (dominant mode: `none`), though the pair consistently ranks in the upper decile (rank $\leq$ 2,307/8,010) in both batches and both methods. The reverse probe IFN-γ $\to$ IL-12 ranks higher under the refined method (786/8,010 in Batch 1), reflecting the fact that the refined score separates the two directions independently rather than conflating them into a single antisymmetric value. IL-10 $\to$ IL-6 is similarly elevated (rank 722, Batch 1 refined), consistent with IL-10's known suppressive feedback on IL-6 signalling.

Figure 5 shows the cross-batch scatter for both methods.

8 Limitations and Caveats

1. **Encoder is cell-type supervised, not cytokine supervised.** Cytokine signals are encoded incidentally, as deviations within cell-type clusters. If cytokine A induces a program identical to resting-state variation in cell type T, it will not appear as a displacement in PBS-RC space.
2. **Small effective donor sample.** All centroids are computed from $|\mathcal{D}_{\text{train}}| = 10$ donors. Pseudo-tubes from the same donor are correlated; effective $n = 10$ donors, not the number of

cells. The Wilcoxon test in the refined pipeline operates on this donor-level vector, which is the correct unit of analysis, but statistical power is limited.

3. **24-hour snapshot.** All tubes are measured at a single 24-hour time point. Cytokines with slow induction kinetics may not yet have produced sufficient secondary cytokine B for the effect to appear. A positive score is consistent with a cascade but could also reflect shared transcriptional programmes activated simultaneously.
4. **Top-50 overlap remains low.** Even under the refined method, top-50 cross-batch overlap reaches only 10/50 (Table 3, row 2). This indicates a noise floor at the level of individual pair rankings. The monotone improvement with filtering suggests a threshold problem rather than absent signal, but stricter BH thresholds or more seeds are needed to verify.

9 Conclusions

1. **Within-batch stability is high.** The legacy method achieves 10/12 seeds stable in Batch 0 and 11/12 in Batch 1; the refined method achieves 11/12 in both.
2. **Cross-batch reproducibility clears the pre-specified threshold.** Legacy $\rho = 0.766$, refined $\rho = 0.815$, both exceeding the pre-specified 0.70 threshold on disjoint training data.
3. **Legacy high-magnitude pairs are its least reproducible.** Restricting to the top 50% of pairs by legacy magnitude drops legacy ρ to 0.620 while raising refined ρ to 0.867 on the same pairs, exposing contamination as the dominant driver of large legacy scores.
4. **The refined method's signal concentrates toward the top.** Refined ρ improves monotonically with filtering, demonstrating that the method carries genuine biological signal rather than noise.
5. **Pre-registered controls behave as expected.** IL-12 $\rightarrow$ IFN-γ ranks consistently in the upper decile with correct directionality; the IL-6/IL-10 negative control is mid-ranked and approximately symmetric.
6. **IL-27 and IFN-β emerge as robust cascade sources.** Their targets overlap substantially between the two independent batches and match documented downstream biology.

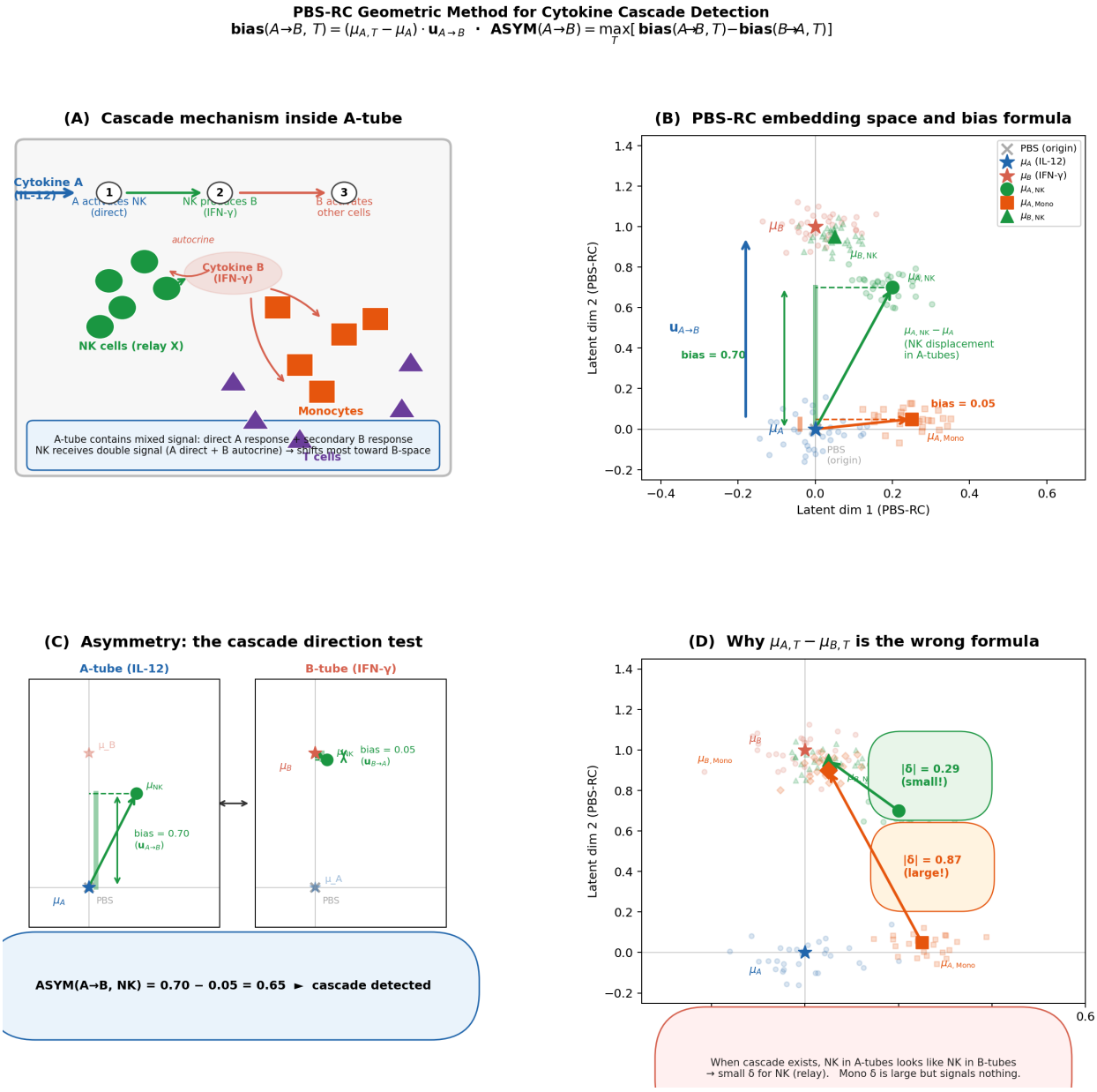

Figure 2: **PBS-RC geometric method for cascade detection.** **(A)** Biological mechanism: cytokine A directly activates relay cell type X (NK cells), which secrete B autocrinally and into the tube paracrinally. **(B)** PBS-RC embedding space: NK cells in A-tubes are strongly displaced toward B (bias $= 0.70$); Monocytes barely deviate (bias $= 0.05$). **(C)** Asymmetry test: large forward bias minus small reverse bias gives $\mathrm{ASYM}(A \to B) = 0.65$, identifying NK as relay X^*. **(D)** Why $\boldsymbol{\mu}_{A,T} - \boldsymbol{\mu}_{B,T}$ is the wrong formula: when a cascade exists, NK in A-tubes resembles NK in B-tubes ($|\delta| = 0.29$, small), while Monocytes differ for unrelated reasons ($|\delta| = 0.87$, large). The naïve formula inverts the ranking.

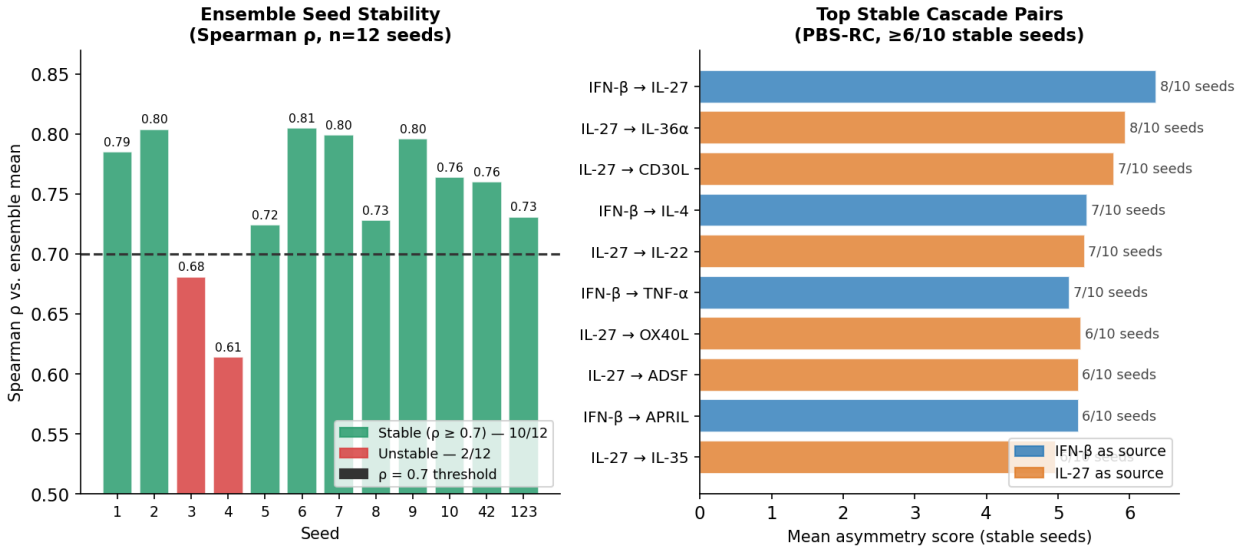

Figure 3: **Legacy method — seed stability and top cascade pairs (Batch 0).** *Left:* Per-seed Spearman ρ versus the batch ensemble mean. 10/12 seeds are stable ($\rho \geq 0.70$); seeds 3 and 4 are unstable (red). *Right:* Top pairs with $\geq 6/10$ stable seeds, ranked by mean legacy asymmetry score. IL-27 (orange) and IFN-β (blue) dominate.

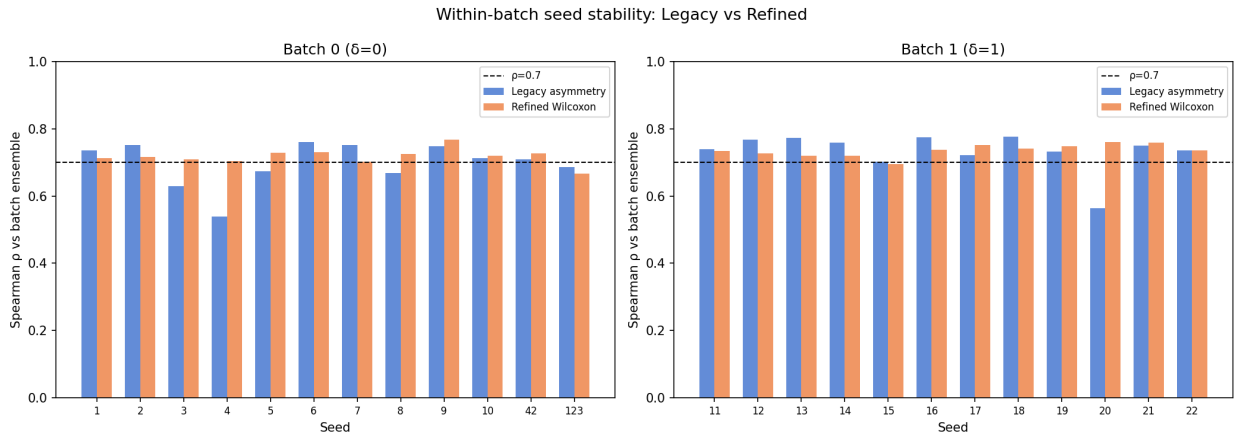

Figure 4: **Within-batch seed stability: Legacy vs. Refined.** Dashed line: $\rho = 0.70$ threshold. Legacy (blue) has five seeds below threshold in Batch 0 (seeds 3, 4, 5, 8, 123); Refined (orange) has one (seed 123 only). Batch 1 is comparable for both methods (11/12 stable each).

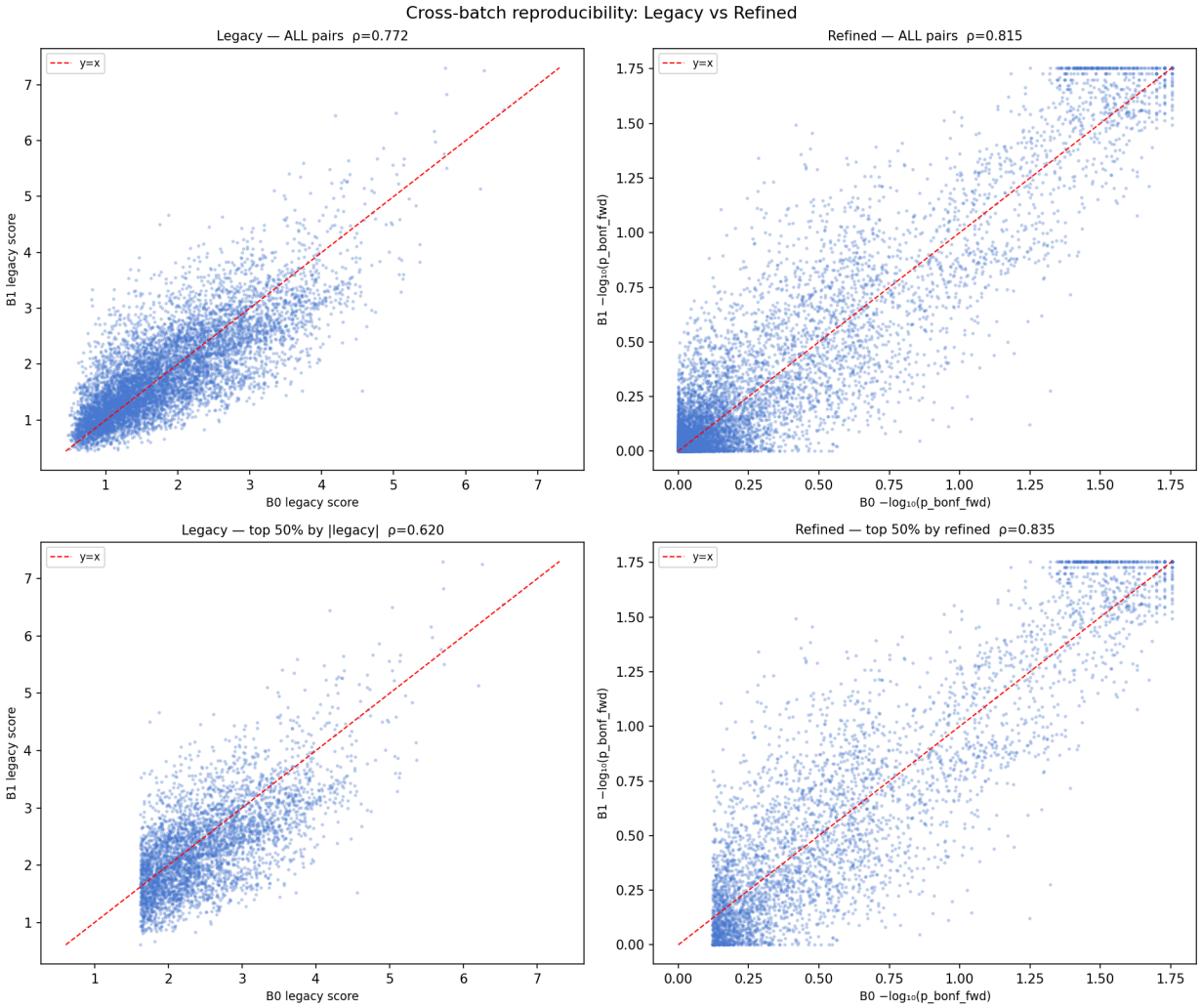

Figure 5: **Cross-batch scatter (Batch 0 score, x-axis; Batch 1 score, y-axis).** *Top row:* all 8,010 pairs. Legacy $\rho = 0.772$ (left); Refined $\rho = 0.815$ (right). *Bottom row:* top 50% by each method's own magnitude. Legacy (left) loses coherence (ρ drops to 0.62); Refined (right) gains it (ρ rises to 0.84). The score ceiling at $\approx$1.75 in the refined plots is a p-value floor artefact ($-\log p$ saturates when all donor Wilcoxon tests are maximally significant).